



School innovation in district context: Comparing traditional public schools and charter schools

Courtney Preston^{a,*}, Ellen Goldring^b, Mark Berends^c, Marisa Cannata^b

^a Vanderbilt University, Peabody 152, 230 Appleton Place, Nashville, TN 37203, United States

^b Vanderbilt University, United States

^c Notre Dame University, 1014 Flanner Hall, Notre Dame, IN 46556, United States

ARTICLE INFO

Article history:

Received 8 January 2011

Received in revised form 13 July 2011

Accepted 29 July 2011

JEL classification:

I21

Keywords:

Charter schools

Innovation

Staffing policies

ABSTRACT

Market reforms in education are part of the educational policy landscape in many countries. Central to arguments for market reforms is the idea that competition and choice will spur changes in schools to be more innovative, which in turn will lead to better student outcomes. We define innovation in terms of a practice's relative prevalence in a local district context. A charter school is innovative in its use of a practice if the traditional public schools in its local school district are not using that practice. We explore factors based on arguments for charter schools that may affect a charter schools' propensity toward innovation to explain variation in levels of innovation across charter schools. We find that, on the whole, charter schools do not fulfill their promise of innovation. Teacher tenure is the most notable exception. Parental involvement is the only characteristic of charter schools that significantly predicts variation in levels of organizational innovativeness.

© 2011 Elsevier Ltd. All rights reserved.

1. Introduction

Central to arguments for market reforms is the idea that competition and choice will spur changes in schools to be more innovative, which in turn will lead to better student outcomes (Chubb & Moe, 1990; Walberg & Bast, 2003). As Lubienski (2003) states, “choice, competition, and innovation are cast as the necessary vehicles for advancing academic outcomes” (p. 397). Despite calls for opening education to the marketplace, we have limited evidence if competition and choice spur innovation. Research supporting or refuting the idea that school choice leads to more innovation is either nonexistent or mixed (Gill, Timpane, Ross, Brewer, & Booker, 2007; Lubienski, 2003).

The purpose of this paper is twofold. First, we compare levels of innovation across two school types: traditional

public schools and charter schools. In defining innovation, we submit that educational practices cannot be deemed innovative in an absolute sense, but innovations must be considered in terms of their relative prevalence in a local and state context. A charter school is innovative in its use of a practice if the traditional public schools in its local school district are not using that practice. Second, we explore factors based on arguments for charter schools that may affect a charter schools' propensity toward innovation and try to explain variation in levels of innovation across charter schools.

1.1. Theoretical claims for innovation in charter schools

Theoretical claims that charter schools spur innovation lie in two realms: the first stems from market theory, the second from the idea that charter schools have greater autonomy and are free from the bureaucracy that may prevent innovation in traditional public schools. Many reformers maintain that market-style mechanisms of consumer choice and competition between autonomous

* Corresponding author. Tel.: +1 615 322 0225; fax: +1 615 322 7890.

E-mail addresses: courtney.preston@vanderbilt.edu (C. Preston), ellen.goldring@vanderbilt.edu (E. Goldring), Mark.berends.3@nd.edu (M. Berends), marisa.cannata@vanderbilt.edu (M. Cannata).

schools will encourage diverse and innovative approaches (e.g., Chubb & Moe, 1990; Walberg & Bast, 2003). Proponents of choice argue that providing this freedom not only diversifies educational opportunities, but also creates incentives for the improvement of traditional public schooling through increased market competition for services. For example, Chubb and Moe (1990) and others (e.g., Friedman, 1962; Lieberman, 1989, 1993) argue that as choice allows market-like competition to increase and bureaucratic structures to decline, it provides parents with greater opportunities to interact with schools and schools with greater openness to parents' demands. Supporters of de-bureaucratization contend that parents, especially low-income and minority parents, will be less intimidated by the schools and more willing to make their needs known to school personnel, resulting in school processes that are more innovative to support students toward higher achievement (Cookson, 2002; Rinehart & Lee, 1991). Based on the supply-and-demand supposition in a system of imperfect competition, we can imagine a situation in which school administrators have almost complete control over the mix of services that they provide and the approaches they use, and a situation in which parents have many choices of schools available for their children (Betts, 2005).

The second perspective that charter schools spur innovation flows from the theory, rooted in charter school legislation, that charter schools are free from bureaucracy and have greater autonomy than traditional public schools (Geske, Davis, & Hingle, 1997; Malloy & Wohlstetter, 2003). According to the Department of Education's Office of Innovation and Improvement, charter schools are "free, in short, to do whatever it takes to build the skills, knowledge, and character traits their students need to succeed in today's world" (2004, 1). Charter schools are designed to be more organizationally independent and more decentralized in terms of their mechanisms of control compared to traditional public schools (e.g., Chubb & Moe, 1990; Raywid, 1995). As Chubb and Moe (1990) point out, "the whole point of a thoroughgoing system of choice is to free the schools from... disabling constraints by sweeping away the old institutions and replacing them with new ones" (p. 219). Charter schools are exempt from many rules that generally govern other public schools, allowing for greater autonomy from both teachers and principals. The assumption is that as school choice undercuts bureaucratic political control of public education, it provides educators in schools of choice the opportunity and motivation to experiment with new organizational and instructional strategies for improving student achievement.

2. Prior research

Findings from research aimed at measuring innovation in charter schools are limited and mixed. An early report on school choice policies in Michigan (Arsen, Plank, & Sykes, 1999) found little difference in charter school practices from those of traditional public schools. They observe innovation in teaching and learning, school organization, and governance and management, implicitly defining innovation as "significant departure from traditional practice" (p. 49). Practices that differed most from those of traditional

public schools were specialized programs of study, small school size, and the basic institutional structure of charter schools, providing autonomy outside the purview of local school boards. Further, conditions like test-based accountability worked against charter schools experimenting with innovative practices.

Another study evaluating Michigan charter schools (Horn & Miron, 2000) found that the practices and procedures in curriculum, instruction, operations, and governance, claimed as innovative by charter schools were often found in local traditional public schools or were well-known practices and procedures among educators. They obtained information from charter schools and their authorizing agencies about practices they considered unique or innovative. Curricular innovations included areas of specialization as well as focus on character and positive school climate, while instructional innovations focused on either teacher- or student-centered pedagogical approaches. Parent and teacher involvement in decision-making like budget and curriculum are key governance innovations. No information is provided concerning how widespread and with what frequency these innovative practices and procedures occur. This report relies on self-report of practices charter schools consider innovative and while it mentions comparing practices in charter schools to those in local traditional public schools, provides no evidence from this comparison group of how frequently these innovative practices occur in traditional public schools.

A similar initial study of autonomy and accountability in Pennsylvania charter schools (Miron & Nelson, 2000), found that most innovations came in teacher and parent involvement in decision-making processes, extended learning time (e.g., Saturday school) and focusing on a specific culture in their curriculum. The Pennsylvania Charter Appeals Board defines innovation as practices or services not provided by the charter school's local school district. While the report mentions a national survey administered to charter schools as a basis for comparison, this traditional public school comparison data is not mentioned in reference to determining innovation. Instead, innovation is determined by charter school differences from perceived traditional practice.

A 2004 report by the U.S. Department of Education's Office of Innovation and Improvement provides case studies of eight successful charter schools from one and two-day site visits. Included in the guides for interviews and observations was the question, "What is innovative about the school's structure and programs?" (p. 4). Among their hallmarks of successful charter schools, they include innovation across the school program, often combining standard practices in unique ways. While not explicitly focused on describing innovation, each portrait of a charter school describes practices that are not conventionally associated with practices found in traditional public schools. However, they rely on perceived conventional practices rather than providing data on actual practices of traditional public schools.

A more recent study of open-enrollment charter schools in Texas notes that applications for open-enrollment charter schools must address how the school plans to encourage innovation (Ausbrooks, Barrett, & Daniel, 2005). These

innovations, often in school mission statements, include plans to help students who have or would have dropped out of traditional public schools and provision of nurturing learning environments. Schools whose mission focuses on dropout prevention were more likely to use innovative instructional practices such as flexible schedules and self-paced learning, while other charter schools are more similar to traditional public schools. Unfortunately, while Ausbrooks et al. provide percentages of charter schools in their studies employing these practices, they do not provide percentages of traditional public schools employing the same practices for comparison. Even so, they conclude by noting “there are few charter schools in Texas that are truly creating innovative teaching environments” (p. 14).

A report on Texas charter schools from the same year makes strong claims that they are innovative, but provides little support for this claim (Smith, 2005). The report names four areas of innovation, preparation for higher education, teamwork, student assessment, and funding sources. In each category, the report provides one anecdote of an innovative charter school, with little conceptualization of innovation or provision of context for these innovations.

Berends, Goldring, Stein, & Cravens (2010) analyze the relationship between four levels of instructional conditions and student mathematics achievement using a propensity score matched sample from schools in Indiana, Idaho, and Minnesota. They measure academic instructional innovation based on teachers’ perceptions of school improvement efforts and focus on innovations mostly closely related to student learning (e.g., whether instructional program was unique and based on research evidence). They find no significant relationship between instructional conditions as a whole and student math gains, however, they do find a strong negative relationship between the innovation and math achievement, findings that are robust to controls at the state, school, and student levels.

Each of the previous studies reports on innovation in charter schools in one particular state or focuses on individual charter schools. They provide varying definitions of innovation and none provides a comparison group of traditional public schools for measuring innovative practices against, instead relying on perceptions of common practices in traditional public schools, both locally and on the whole. Contrasting these school- and state-level investigations of innovation, the National Charter School Research Project (2007) measures innovation in charter and traditional public schools nationally, using data from the National Center for Education Statistics’ 1999–2000 Schools and Staffing Survey. They find charter schools are more likely to use enrichment programs, team teaching, and block scheduling as well as special instructional approaches like Montessori programs. While this study does measure innovation in charter schools against a clear comparison group, they measure innovation in the aggregate, with no allowance for innovation in a local context.

3. Conceptualizing innovation

Beyond the theoretical arguments that charter schools should spur innovation, there is little agreement as to what is considered innovative in the school choice debates. Even

the definition of innovation is varied. What is considered innovative for one may be standard or conventional practice for another. Some have claimed that charter schools in and of themselves constitute an innovation, in that they are a new form of governance for publicly funded schools and create a new institutional arrangement that may change the way schools cooperate with and compete against one another (Center for Education Reform, 2008; U.S. Department of Education, 2008; Bulkley & Fisler, 2003; Davis & North, 1970). Others view charter schools as the research and design centers for education, where truly brand new practices in education are developed, implemented, and ultimately diffused to other schools (Lubienski, 2004; Miron & Nelson, 2000). For some charter schools innovation may simply mean the ability to hire their own teachers, whereas other charter schools may be working toward implementing comprehensive school instructional designs that aim to change the teaching staff, their professional development, the organization of instructional time, curriculum, and instruction (Berends, Cannata, Goldring, & Penaloza, 2009).

In general, innovation is based on the perception of a practice or process as being new (Grunert et al., 1997; Kotler, 1991; Rogers, 1983; Schumpeter, 1934). Schumpeter (1934) defined economic innovation as the introduction of a good with which consumers are not familiar into a market. Thus, a practice that is innovative in one context may not be innovative in another. In defining innovativeness, a distinction must be drawn between innovativeness as a characteristic of an organization and innovativeness as a characteristic of an innovative practice or product (Grunert et al., 1997). Innovation as a characteristic of a product is determined by how new that product is to its market, while innovativeness as an organizational characteristic is a function of the organization’s quality of innovation, capacity and willingness to innovate, and speed of innovation (Cozijnsen & Vrakking, 1993). In other words, organizational innovativeness is a function of organizational culture and climate. Mowery and Rosenberg (2000) further elaborate, noting that innovation must include local structures and dynamics, considering the context of the innovation. Therefore, while innovative practices may not be prevalent or standard practice in traditional public schools, neither are they innovative in the “brand new” sense of the term: they are different from business as usual in neighboring traditional public schools. We first consider innovativeness as a characteristic of a practice and then turn to predicting charter schools’ organizational innovativeness.

3.1. Innovation defined in charter schools

In the field of education, Lubienski (2003) conceptualizes innovation in terms of “educational changes, (practices regarding curricular content and instructional strategies with immediate impact at the classroom-level), and administrative changes (organizational-level practices and structural designs that do not directly affect classroom techniques or content)” (pp. 404–5, emphasis in original). Similarly, Berends and King (1994) classified school restructuring based on four themes or dimensions:

student experiences, professional lives of teachers, school governance, management and leadership, and community resources. Following these two frameworks and the arguments for school choice, we conceptualize innovation in terms of student experiences defined in the areas of academic support services and organizational restructuring for students, and administrative innovations in the areas of teacher compensation and governance. Both the arguments for choice in terms of increased autonomy for educators and competition theory predict that charter schools would exhibit innovations in terms of offering needed academic programs to students. The market would demand them, in the form of parent choice and pressure, and teachers and administrators would be free from bureaucracy and restricted budgets to offer them. Academic support services in the form of afterschool tutoring, extended year schooling, and learning programs such as distance learning and language immersion are such educational innovations that can enhance the academic achievement of at-risk students.

An additional aspect of enhancing the educational experiences for students pertains to how students are grouped. Schools have altered conventional instructional designs to emphasize alternative instructional grouping arrangements. Because research has shown that students placed in high ability groups tend to learn at a greater rate than students placed in low ability groups—contributing to greater inequality in achievement scores (for a review see Gamoran & Berends, 1987; Gamoran & Hallinan, 1995; Oakes, Gamoran, & Page, 1992)—some have argued for detracking and alternative grouping arrangements (Welner & Oakes, 2000; Wheelock, 1992). Some of the innovations around alternative grouping arrangements include flexible schedules, “looping” (where teachers progress with the same group of students across grade levels), or multi-grade or mixed-age grouping (see examples in Glennan, Bodilly, Galegher, & Kerr, 2004).

In terms of administrative innovations, one of the most widespread arguments for school choice is in regard to the notion that charter schools would be free from rules and regulations that restrict the teacher labor market. Podgursky (2008) reports that teachers at charter schools are more likely to have incentive pay and bonuses to recruit and retain them than traditional public school teachers and are less likely to have their salaries determined by a single salary schedule. In addition, teacher dismissal rates are higher in charter schools, where teachers are less likely to be covered by collective bargaining agreements (Podgursky, 2008). Charter schools also have greater leeway in hiring practices. Teachers at charter schools are much less likely to be union members, and therefore are not part of collective bargaining agreements and tend to be hired directly by the individual charter schools, rather than a centralized agency like a school district (Bulkley & Fisler, 2003; Koppich, Holmes, & Plecki, 1998). The implication of these hiring practices is that charter schools have more flexibility in their working (contractual) relationships with teachers (Malloy & Wohlstetter, 2003; Podgursky & Ballou, 2001).

Another area where charter school teaching should be more innovative is in the area of school governance,

specifically teachers' influence on school decisions. School governance is touted as a primary arena for innovation given the autonomy afforded charter schools by state legislation (Arsen et al., 1999; Bulkley & Fisler, 2003; Lubienski, 2003; Miron & Nelson, 2000). Bifulco and Ladd (2005) found higher levels of school autonomy in charter schools, such as influence over school programs and policies. Teachers in charter schools report greater influence over academic standards and curriculum compared to their counterparts in traditional public schools (Podgursky, 2008). Cannata (2008) similarly found that charter school teachers are more likely than their peers at traditional public schools to indicate involvement in school governance and having influence on school policies.

3.2. Research questions

Following our conceptualization of innovation in the local context, we ask,

1. Do levels and types of innovation differ between charter schools and traditional public schools taking into account local district context?
2. What factors predict a charter schools' propensity toward innovation?

In this paper, our framework measures innovation in terms of how innovative a practice is relative to its local district context.

4. Method

4.1. Sample

To study differences in innovation between charter schools and traditional public schools, we use the 2007–08 Schools and Staffing Survey (SASS) (NCES). The SASS is a nationally representative sample which provides a picture of practices in use at both the school and district level. Because we define practices as innovative relative to their local context, we chose SASS to provide this context, as its survey universe includes both charter and traditional public schools.

SASS is administered by the National Center for Education Statistics (NCES) and provides a nationally representative sample of teachers and schools, pulled from the 2005–06 Common Core of Data (CCD) sampling frame. We use data from the school district, school administrator, and school surveys. Our sample is limited to charter schools and traditional public schools. To provide local context for charter schools in order to determine the extent to which they are innovative in relation to traditional public schools in their locales, we linked charter schools to traditional public schools in the same district using NCES district identifiers.

To link charter schools to traditional public schools in whose districts they are situated, we used a two-step process. First, we created a new identifier for each charter school in SASS. In some states, charter schools are assigned the same NCES district identifier as the school district that authorized them. For these charter schools, we used NCES district identifiers to determine which traditional public

schools were in same school district and then assigned the same identifier to both charter schools and the traditional public schools in a given district. In other states, regardless of how the charter school was authorized, the charter school is its own district and is assigned a unique NCES district ID. In order to link such charter schools to traditional public schools in the school district they are located in, we located each charter school's physical address on a map and determined which school district it fell within the geographical boundaries of.¹ We then assigned these charter schools and the traditional public schools in the district where they are located the same identifier. Because of the SASS's sampling scheme, the survey universe does not include both traditional public schools and charter schools for each district. As such, when we linked charter schools to traditional public schools in this manner, we created a final sample of 203 charter schools and 739 traditional public schools in 36 states.

In order to compare schools at the elementary, middle, and high school levels, we created an identifier for each level. A school is identified as an elementary school if it contains any of grades Kindergarten through fifth grade, identified as a middle school if it includes any of grades six through eight, and identified as a high school if it includes any of grades nine through twelve.

4.2. Data

Our data come from the district, school, and principal questionnaires from the 2007–08 Schools and Staffing Survey. SASS questionnaires focus on teacher and administrator characteristics, school programs, and school conditions. The SASS public school district survey had a response rate of 87.8%, the public school survey response rate was 80.4%, and the public school principal response rate was 79.4%. SASS does not differentiate between charter schools and traditional public schools in reporting these figures.

4.3. Measures

We chose items from the SASS surveys that measure the implementation of innovative structures and practices. Following arguments for innovation in charter schools (Allen, 2001; Betts & Loveless, 2005; Budde, 1988; Bulkley & Fisler, 2003; Hess & Lovelace, 2005), we organized measures of innovation from SASS into four broad categories of innovation: (1) staffing policies; (2) academic support services; (3) school organizational structures; and (4) governance (see Appendix A).

Staffing policies: Given that charter schools are more likely to be able to hire teachers that are not governed by unions the area of teacher compensation and tenure is a key area for innovation. Questions around teacher pay structures include whether a school or district uses pay

incentives to reward excellence in teaching and improved student achievement or to recruit and retain teachers in shortage fields. We include whether or not a school offers its teachers tenure in this category.

Academic support services: A second area of innovation pertains to academic support services. Many charter schools are actually founded to help at-risk students succeed. Academic support services measures whether schools offer voluntary or mandatory after school tutoring or extended day instruction for students in need of academic assistance, and whether schools offer summer school or tutorial programs for students. This area also includes program offerings including foreign language immersion programs, distance learning programs, and internships.

Organizational structures: Charter schools are argued to be more flexible and innovative in their school organizational structures. Our measures of instructional organization include whether schools are year round and whether they use block scheduling, looping, “houses” or “families” that are larger than one classroom, and multi-grade or mixed-age classrooms. We include schools requiring a standardized dress code or uniforms as an organizational policy as well.

Governance: Furthermore, charter school autonomy and stakeholder involvement in decision-making are a hallmark of charter school advocates. Measures of stakeholders' influence on hiring new staff come from the question “How much ACTUAL influence do you think each group or person has on decisions concerning the following activities? Hiring new full-time teachers of this school” (NCES, 2008). Ordinal answer choices for each stakeholder (teachers, parents, and principals) are “no influence,” “minor influence,” “moderate influence,” and “major influence.” For each stakeholder, we created binary variables where responses of “no influence” or “minor influence” equal zero and responses of “moderate influence” or “major influence” equal one. We use the influence of teachers and parents in our analysis (see Appendix A for specific items from the SASS surveys).

4.4. Analyses

In order to determine whether a charter school is innovative, we tabulated each charter school's response to each item alongside its matched traditional public schools in the same district. For each charter school on each item of innovation, we then categorized the responses into one of the following scenarios:

- (1) A charter school reports having the practice and none of the traditional public schools report having the practice. In this case, the charter school is innovative. Or, conversely, with some measures of innovation (e.g., tenure) a charter school does not report having a practice, while all of the traditional public schools in the same district do, in this case the charter school is innovative.
- (2) A charter school and all of the traditional schools in the district report having the practice or all report not

¹ Our approach is limited in that we do not know if these charter schools are drawing students from the geographical districts in which they are located. Without student level data, we cannot address this issue. However, as we consider innovation in its local context, considering the geographical boundaries of school districts is appropriate.

having the practice. In these scenarios, none of the schools are innovative.

- (3) Finally, a charter school reports having a practice and some, but not all of the traditional public schools report having the practice as well. In this case, we use a 20% “tipping point” decision rule (Rogers, 1995) to determine whether or not a charter school is innovative. If 20% or more of the traditional public schools in a district report having the same practice that a charter school reports having (or does not report a practice that a charter school does not report having), then that charter school is not considered innovative.²

Our first analysis describes the overall level of innovativeness of charter schools compared to traditional public schools in the districts the schools are situated in. These measures are cross-sectional, identifying practices at only one point in time. As such, in a district where both the charter school and traditional public schools employ a practice, we are unable to identify wherein the practice originated and its path of diffusion.

To answer our second research question, what factors predict a charter schools’ propensity toward innovation, we create ordinal outcome measures that are a count of the number of innovative practices a charter school employs in each of our four categories (presented in the results section).³ As these outcome measures have multiple discrete categories and measure a latent level of innovation, we then estimate ordered probit models to estimate the influences of particular characteristics on a charter school’s innovation in each of our four categories of innovation.⁴

These models take the general form

$$Y_i = \beta_0 + \beta_1 \text{ enrollment} + \beta_2 \text{ achievement} \\ + \beta_3 \text{ principal background} \\ + \beta_4 \text{ parental involvement} + \beta_5 \text{ turbulence} + e_i,$$

where Y_i has as many ordered discrete outcomes as the number of innovative practices a charter school implements in each category: academic support services, staffing policies, organizational structures, and governance. *Enrollment*, measured in hundreds, is the school’s enrollment as reported by the principal on SASS. We would expect smaller charter schools to be more innovative because of the autonomy and freedom from bureaucracy afforded charter schools in general and the flexibility and ease of implementation of innovation in smaller charter schools in particular (Bulkley & Fisler, 2003; Griffin & Wohlstetter,

2001; Lubienski, 2003; Miron & Nelson, 2000). While not an innovation we measure, small school size is cited as a innovation in charter school organization (Arsen et al., 1999).

Achievement is a binary variable equal to one if the school has been designated for improvement, as indicated by failing to make Adequate Yearly Progress for two consecutive years, 2005–06 and 2006–07 and zero otherwise. Inherent in the premise of charter schools is that they will lead to increased student achievement in return for the increased autonomy they are afforded (Bulkley & Fisler, 2003; Griffin & Wohlstetter, 2001; Miron & Nelson, 2000). This increased autonomy should enable teachers and administrators to test new strategies and approaches for increasing student achievement (Allen, 2001; Budde, 1988; Goldring & Cravens, 2006). While this is a rough measure of academic achievement, it provides evidence of a school’s trend in achievement. We would expect that whether a charter school is identified for improvement for not making AYP for two consecutive years would influence their level of organizational innovation, although the direction is not clear. They may experiment with unique grouping strategies, organizational structures, and governance mechanisms to improve student achievement and reach accountability goals or conversely return to a focus on core curriculum.

School leadership has been shown to have effects on student achievement and schools’ capacity for effective teaching and learning (Fullan, 2007; Hallinger & Heck, 1996; Leithwood & Jantzi, 1999). As such, we include principal background characteristics as predictors of a charter school’s propensity toward innovation. *Principal background* is a vector of binary variables measuring a principal’s years tenure in the charter school and management experience outside of the field of education. We created three mutually exclusive binary variables to measure a principal’s years of experience in the current school: first year in the school, two to five years in the school, and more than five years, with more than five years the excluded category. We hypothesize that while in a principal’s first year in a school they are less likely to make changes, most innovation will occur early in their principalship. Triant (2001) provides evidence that while a principal may experiment with innovation in the early years of their tenure in a school, after a few years, test-based accountability may force them to return to business as usual to accommodate state curriculum standards. Findings related to charter school leadership experiences outside of education are mixed. Some studies report few charter school leaders having management experience at all (e.g., Finn, Manno, & Vanourek, 2000; Griffin & Wohlstetter, 2001) while others report that charter school leaders are more likely to have backgrounds outside of education than leaders in traditional public schools (Bulkley & Fisler, 2003). Management experience is equal to one if a principal responded yes to the question, “Before you became principal, did you have any management experience outside of the field of education?” and zero otherwise. We hypothesize that principals with management experience outside of education will be more likely to innovate as they bring their varied experience to bear in the school.

² The same process could certainly be used to determine whether traditional public schools are innovative. However, this is beyond the scope of this study.

³ We include only charter schools as our research questions focus only on innovation in charter schools.

⁴ We use an ordered probit model instead of a multinomial probit model because, for each of our outcome measures, a higher score does mean a greater amount of innovation. Further, our outcomes do not satisfy the Independence of Irrelevant Alternative assumption require for multinomial probit models, while the ordered probit model relaxes this assumption.

The theory behind charter schools is that parents will bring pressure to bear on schools to be innovative. Prior research also evidences a high level of parental involvement in charter schools compared to traditional public schools (Bulkley & Fidler, 2003; Miron & Nelson, 2000). We predict that higher levels of parental involvement will be associated with the propensity to be innovative in charter schools. Parents who choose to put their students in charter schools may be looking for an educational experience outside of what traditional public schools can provide, expecting innovation in the charter school they choose. Our measure is a four-item scale ($\alpha = 0.88$), created from principal quartile responses to, “LAST school year (2006–07), what percentage of students had at least one parent or guardian participating in the following events: (a) Open house or back-to-school night; (b) All regularly scheduled schoolwide parent–teacher conferences; (c) One or more special subject-area events (e.g., science fair, concerts); (d) Volunteer in the school on a regular basis.” Supportive and involved parents have been found to impact charter schools’ success in creating and sustaining learning communities (Wohlstetter & Griffin, 1998).

Charter schools often face turbulent environments, as start up organizations. We measure *turbulence* by difficulty in recruiting teachers based on the idea that a charter school with a turbulent environment, operationalized by difficulty recruiting teachers, may be more likely to innovate and try new ways of improving student achievement, rather than being able to rely on high quality teachers. This measure is a 12-item scale ($\alpha = 0.85$), where each item is a principal’s four point Likert-scale response to the question “How easy or difficult was it to fill the vacancies for this school year in the following fields?” for a different teaching field with 1 being “Easy” and 4 being “Could not fill the vacancy.”

Additionally, because of the strong association between student background characteristics and student achievement, we control for student minority status and students in poverty, as measured by the proportion of students eligible for free and reduced price lunch (Hanushek, 1997).

Table 2

Distribution of innovative practices in charter and traditional public schools.

Innovation practice	Charter schools reporting practice	Traditional public schools reporting practice	Innovative charter schools
<i>Academic support services</i>			
Voluntary tutoring	58.8%	57.6%	6.8%
Summer school	100%	100%	0%
Language immersion program	4.9%	7.3%	3.7%
Distance learning program	17.9%	12.0%	13.6%
Internships	47.4%	70.7%	7.0%
<i>Staffing policies</i>			
Merit pay	32.1%	18.7%	13.1%
Tenure	8%	100%	92%
Shortage fields pay	40.7%	59.6%	3.3%
<i>Organizational structures</i>			
Looping	35.2%	20.7%	17.3%
Block scheduling	47.8%	43.3%	15.5%
Year round schedule	19.3%	24.5%	7.5%
Houses	21.1%	23.3%	9.3%
Mixed age/multi-grade grouping	39.8%	28.1%	16.8%
<i>Governance</i>			
Teachers have influence on new staff hiring	71.1%	72.3%	6.0%
Parents have influence on new staff hiring	22.2%	18.1%	14.5%

Table 1

Descriptive statistics of innovativeness predictors.

	Mean	Standard deviation
Enrollment (hundreds)	3.927	4.543
AYP status	0.661	0.474
Management experience	0.546	0.499
Principal tenure in school	3.430	3.629
Teacher recruitment difficulty	3.824	0.601
Parental involvement	2.767	0.898
Proportion minority	0.504	0.374
Proportion eligible for free/reduced price lunch	0.454	0.344

Descriptive statistics for these covariates in the sample of charter schools are provided in Table 1.

5. Results

On the whole, we find that charter schools in our sample are not more innovative than traditional public schools. The results are summarized in Table 2. Column 1 shows the frequency with which an innovation is found in our entire sample of charter schools, column 2 shows the frequency with which they are found in our sample of traditional public schools, and column 3 show the percentage of charter schools that are innovative in their use of a practice in their local context. Given the stability of the education system over several decades, we consider charter schools innovative in a practice when more than 10% of charter schools in our sample are innovative in their use of a practice.

In addition, we consider the possibility that innovation varies by grade level. Use of one practice may not be innovative when implemented at the high school level, but its use at the elementary level may be innovative. As such, we determine whether charter schools are innovative in their practices as compared to other schools serving the same grade levels. Table 3 provides these comparisons. Elementary and middle schools demonstrate more innovation than high schools and charter schools in the aggregate, particularly in the areas of organizational structures.

Table 3

Distribution of innovative practices in charter and traditional public schools by school grade level.

Innovation practice	Innovative charter schools: elementary level	Innovative charter schools: middle school level	Innovative charter schools: high school level
<i>Academic support services</i>			
Voluntary tutoring	12.0%	14.6%	9.4%
Summer school	0%	0%	0%
Language immersion program	6.1%	4.8%	0%
Distance learning program	12.2%	17.9%	13.2%
Internships	0%	23.5%	5.7%
<i>Staffing policies</i>			
Merit pay	16.0%	15.4%	8.3%
Tenure	92.1%	89.8%	91.5%
Shortage fields pay	4.0%	3.8%	5.7%
<i>Organizational structures</i>			
Looping	22.2%	16.9%	13.5%
Block scheduling	22.2%	14.3%	11.3%
Year round schedule	12.2%	11.0%	7.5%
Houses	13.9%	11.9%	9.4%
Mixed age/multi-grade grouping	26.8%	25.0%	13.5%
<i>Governance</i>			
Teachers have influence on new staff hiring	9.1%	8.9%	1.9%
Parents have influence on new staff hiring	16.4%	12.5%	10.5%

5.1. Academic support services

First, we turn to innovations in regard to academic support services such as summer school, tutoring and the like. Of the charter schools in our sample, 100% offer summer school programs, as do 100% of the district schools, making none of the charter schools innovative in this practice. In the case of afterschool tutoring, 57.6% of traditional public schools offer this support service for their students compared with 58.8% of charter schools. Of the 94 charters schools offering voluntary tutoring, 11 are innovative in their local context, 6.8% of the total charter sample. That is, while similar percentages of charter schools and traditional public schools offer afterschool tutoring, there are a few who are innovative in their context because they are located in areas where tutoring is not as readily available in traditional public schools. However, when considering charter schools by grade level, we find more charter schools who are innovative in their local contexts: while only 9.4% of charter high schools offering voluntary tutoring are innovative, 12.0% of elementary charters and 14.6% of middle school charters are innovative in their local context.

A much lower percentage of charter schools (47.4%) offer their students internships than traditional public schools (70.4%). In spite of the greater prevalence of work-based learning in traditional public schools in the aggregate, there are 4 charter schools (7.0%) where offering this service is innovative. Internships appear to be a practice widely used in public schools, leaving little room for innovation when charter schools do offer them. However, 23.5% of charter schools offering internships at the middle school level are innovative. Few schools across sectors offer foreign language immersion programs: 4.9% of charter schools sampled and 7.3% of traditional public schools. Six of eight charter schools offering language immersion programs (3.7% of the total sample) are innovative in their local context. Although few charter schools overall offer language immersion programs, when they do offer them, this offering is usually innovative in its context. Similarly, relatively few schools across sectors offer distance learning

programs. Twenty-nine of 162 charter schools (17.9%) and 89 of 739 traditional public schools (12.0%) offer such a program. Twenty-two of these charter schools (13.6% of the total sample) are innovative in this practice. Distance learning programs are the only practice in the area of academic support services where charter schools are innovative in the aggregate. When considering charter schools at the three different levels, charter middle schools implementing these academic support services show more evidence of innovation in their local contexts. On the whole, we find that, with the exception of distance learning programs, charter schools are not innovative in their offerings of academic support services.

5.2. Staffing policies

Next we turn to innovation related to staffing policies. Fourteen of the charter schools in our sample do offer tenure to their teachers. In the case of tenure, not engaging in the practice is considered innovative. Therefore of the charter schools in our sample, 92%, are innovative in that they do not offer their teachers tenure. Forty-four out of 137 charter schools (32.1%) report awarding teachers extra pay for excellence in teaching, while only 127 of 680 traditional public schools (18.7%) do. Eighteen of these charter schools (13.1%) are innovative in their use of merit pay, as none of the traditional public schools in their districts report using merit pay. Considering innovation at differing school levels, high schools demonstrate less innovation in offering merit pay, wherein only 8.3% of high schools are innovative in this practice. Of the 150 charter schools responding to whether or not they offer extra compensation to teachers in fields of shortage like math and science, 61 schools (40.7%) report that they do. In traditional public schools, 406 out of 681 responding schools (59.6%) offer such compensation. Given the prevalence in traditional public schools of extra compensation for teachers in shortage fields, only 5 charter schools (3.3%) are innovative in their local context. While charter schools display some innovation in their use of merit pay, teacher tenure practices appear to

be the area where charter schools are most likely to be innovative.

5.3. Organizational structures

The third area of innovation we examine is innovation regarding organizational structures. Relatively few schools use looping as an organizational practice, although the practice is more prevalent in charter schools than traditional public schools. Fifty-seven out of 162 charters (35.2%) and 153 of 739 traditional public schools (20.7%) report employing this grouping strategy. Of the charter schools using looping, about half of them are innovative in their local context, 17.3% of the total charter sample. Disaggregating charter schools by grade level, we find that elementary charters are more innovative in their use of looping (22.2%). Using houses or families to group students has similar prevalence in charter and traditional public schools. Thirty-four out of 161 charter schools (21.1%) report using houses to organize their students, while 172 out of 739 traditional public schools (23.3%) use houses. These percentages are very similar and only fifteen of the charter schools (9.3%) are innovative in their context. About forty percent (64 of 161) charter schools in our sample report using mixed age or multi-grade classrooms as an instructional grouping strategy, as compared to 28.1% of traditional public schools (208 of 739). Twenty-seven of the charter schools engaging in this practice are innovative, 16.8% of the charter sample. While a smaller percentage of high schools are innovative in this practice (13.5%), greater percentages of elementary charters (26.8%) and middle school charters (25.0%) are innovative in their use of mixed age or multi-grade classrooms, the most innovation we see in organizational structures.

In our sample, 77 out of 161 (47.8%) charter schools report using block scheduling, compared to 320 out of 739 (43.3%) traditional public schools that report using this practice. In spite of the widespread use of block scheduling in traditional public schools, 25 charter schools (15.5%) are innovative in their local context in using block scheduling. In contrast to the widespread use of block scheduling, very few charter schools in our sample employ a year round schedule, only 31 out of 161 (19.3%). The percentage of traditional public schools on a year round schedule is only slightly larger, 24.5% (29 out of 174). However, when considering local context, twelve of these charter schools (7.5%) who use a year round schedule are innovative.

5.4. Governance

Lastly, we turn to governance. Of the 147 responding charter school principals in our sample, 106 (71.1%) report that teachers have influence over new staff hires, while in traditional public schools, 72.3% report that teachers have influence over new staff hires. In spite of the similarity in overall proportions of schools that engage in this practice, there are nine charter schools where teachers having influence over new staff hires is an innovative practice in context. In this case, only 6.0% of charter schools are

innovative in their local context in teachers having influence over new staff hires.

Twenty-six out of 117 responding charter school principals, 22.2%, report parents having an influence over new staff hires, whereas 18.1% of traditional public schools report that parents have influence on new staff hiring. There are 17 charter schools (14.5%) where parental influence on new staff hiring is innovative in its local context.

In sum, we find that on the whole, charter schools do not fully fulfill their promise of innovation. Further, when we examine charter schools in their local contexts to measure their innovativeness, we still find little evidence of innovation. Student grouping structures and staffing policies are the primary areas in which charter schools are innovative, particularly in their offering of tenure and merit pay and use of block scheduling, looping and mixed age/multi-grade classrooms.

5.5. Predicting innovation

Despite the relatively low percentages of charter schools as a group that are innovative in their use of particular practices, there are some charter schools that are innovating in these areas. This section turns to our second research question to explore what predicts why some charter schools may be innovative. We summarize the overall levels of innovation in charter schools in our sample by computing a total score for each type of innovation (see Table 4). We did this by creating a variable for each of our four categories of innovation, equal to the number of innovations in a category that a charter school employs. Governance includes two innovations, so we find that on average, charter schools employ less than one innovative practice. Organizational structures and academic support services both include 5 innovative practices. We learn that overall charter schools are more innovative in terms of academic support services than organizational structures as they implement slightly more than 2 academic support services on average and only one and a half innovative organizational structures. Charter schools are most innovative in teacher compensation which includes three practices.

We next turn to determining which characteristics of charter schools are most likely to predict innovation. Here, we find some evidence that parental involvement and the proportion of students who receive free and reduced price lunch may be related to a charter school's level of innovation as these school characteristics do predict innovation in three of our four categories of innovation.

Table 5 presents the results from our ordered probit models, where each column represents a different category of innovation, based on the count variable (presented in Table 4 above). Considering academic support services, our

Table 4
Descriptive statistics of number of innovative practices.

	Mean	Standard deviation	Range
Academic support services	2.091	1.525	0,5
Staffing policies	1.653	0.713	0,3
Organizational structures	1.532	1.141	0,4
Governance	0.815	0.674	0,2

Table 5

Charter school characteristics predicting innovation.

	Academic support services	Organizational structures	Teacher compensation	Governance
Enrollment (100s)	0.027 (0.019)	−0.010 (0.017)	0.006 (0.018)	0.013 (0.018)
Needs improvement	−0.018 (0.193)	0.004 (0.193)	0.118 (0.211)	0.333 (0.212)
First year principal	0.247 (0.236)	−0.150 (0.234)	−0.187 (0.259)	−0.065 (0.253)
2nd–5th year principal	0.064 (0.186)	−0.315* (0.186)	−0.137 (0.201)	−0.295 (0.201)
Management experience	0.271* (0.163)	0.215 (0.163)	−0.135 (0.177)	−0.046 (0.176)
Parent involvement	−0.189** (0.096)	0.031 (0.094)	0.192* (0.104)	0.439*** (0.107)
Difficulty recruiting teachers	−0.017 (0.131)	0.111 (0.130)	−0.011 (0.142)	−0.148 (0.141)
Minority	−0.017 (0.261)	−0.155 (0.258)	0.227 (0.285)	−0.143 (0.279)
Free/reduced price lunch	0.707 (0.297)	0.223 (0.290)	0.055 (0.317)	0.662** (0.319)
κ_1	−1.168 (0.644)	−0.334 (0.635)	0.503 (0.693)	0.194 (0.686)
κ_2	−0.290 (0.641)	0.385 (0.640)	1.587 (0.701)	1.835 (0.696)
κ_3	0.720 (0.645)	1.258 (0.640)		
κ_4	1.038 (0.646)	2.169 (0.650)		
N = 203				

Note: Standard errors in parentheses.

* indicates statistical significance at .01 level

** indicates statistical significance at .05 level

*** indicates statistical significance at .001 level

findings indicate that increased levels of parental involvement predict a decrease in a charter school's level of innovation in academic support services. It is possible that increased levels of parental involvement signal less need for academic support services provided by the school; parents are providing this support for their students. Interestingly, A principal with management experience prior to entering education predicts an increase in the level of innovation in academic support services. The other characteristics we hypothesized to predict a charter school's level of innovation in this area do not have a relationship to its innovative academic support services. Turning to the area of organizational structures, we find little evidence that any of our characteristics are significantly related to a charter school's level of innovation. The relationship between having a principal with two to five years experience and innovation in organizational structures has marginal statistical significance, indicating a negative relationship between having such a principal and a charter school's level of innovation. However, in both the areas of teacher compensation and governance, we find a positive relationship between parental involvement and the level of innovation in a charter school. A unit increase in the level of parental involvement predicts a 0.192 increase in the likelihood of a charter school being innovative in our latent teacher compensation innovation measure and a 0.439 increase in the likelihood of charter school being innovative in our latent innovation in governance measure. This is about one-fifth of a threshold level for innovation each category, a small relationship.

6. Discussion

Proponents of school choice argue that the autonomy afforded charter schools facilitates innovation and will lead to increased student achievement (Finn, Manno, & Vanourek, 2000; Wohlstetter & Griffin, 1998; Zimmer & Buddin, 2007). This increased autonomy in areas of control over discipline, curriculum, and staff management has not been found to lead to increases in student test scores (Zimmer & Buddin, 2007). We find that this increased

autonomy does not lead to what Lubienski (2003) characterizes as administrative innovation in charter schools on the whole either. The exceptions to this are in student grouping strategies, such as looping and mixed age or multi-grade classrooms, and, most notably, tenure.

In describing innovation in charter schools, a number of studies (e.g., Miron & Nelson, 2000; Triant, 2001) note that while charter schools employ practices that are conventionally used in traditional public schools, it may be the unique combinations of these practices in charter schools that constitute innovation. Others note that many charter schools are returning to basic curricula, focusing on conventional instructional practices in order to increase student achievement, while traditional public schools implement every new instructional reform, strategy, and program that comes down the pipeline and are constantly changing their practices (Bulkley & Fisler, 2003). Future research in this vein will explore innovation in practices that are closer to students, such as Lubienski's (2003) educational changes.

Moon (2010) provides a new paradigm conceptualizing such differences in product markets. She observes that many mature product categories are flooded with products and brands where differentiation among them has become meaningless; even as products are marketed as innovative and new, there is no genuine uniqueness between them in the perception of the average consumer. She describes three types of companies and products that are truly innovative or different: (1) those who “defy the augmentation trend in a category in which consumers have come to expect augmentation” (p. 110) like the simplicity of Google, (2) those who provide alternative classifications to those which we are accustomed to (e.g., *The Simpsons* combination of children's cartoon and adult sitcom), and (3) hostile brands whose marketing campaigns flaunt or tout their product shortcomings (e.g., Mini-Cooper, Red Bull). While not innovative in the R&D sense of the term, charter schools that focus on basic curriculum may be innovative in a similar manner to products that defy augmentation and resist implementing every new educational reform posited, innovation by subtraction.

Moon also notes, “nothing generates conformity quite so organically as the existence of a comparative metric” (p. 212). Although charter schools are afforded greater autonomy than traditional public schools, the cost of this autonomy is greater accountability for improving student achievement. There is also evidence in charter schools that while they may initially set out to be innovative, the pressure of test-based accountability leads to them to resort to traditional modes of governance, organization, and academic support services (Griffin and Wohlstetter, 2001; Wohlstetter and Griffin, 1998). In addition to pressures from accountability, this lack of innovation we see in charter schools is consistent with institutional isomorphism, wherein normative expectations lead to conformity and homogeneity (DiMaggio and Powell, 1983; Elmore, 2007; Meyer & Rowan, 1992).

The propensity score analyses revealed that few school characteristics predict greater levels of innovation. As expected, parental involvement was a significant predictor of innovation, indicating that parents may be functioning as the knowledgeable consumers market theory predicts they will. While we hypothesized that whether a school is identified for improvement by or not making AYP for two consecutive years would influence innovation we do not find this to be the case. Charter schools struggling to make AYP and increase student achievement may be those who have taken a back to basics approach, focusing on remediation rather than on innovative practice.

This study provides an initial national analysis of charter school innovation in its local contexts, but there is much analysis yet to be undertaken. Studies of school effectiveness find that practices closest to the student level have the greatest impact on their learning (Gamoran, Nystrand, Berends, & LePore, 1995; Newmann, 1996; Newmann & Wehlage, 1995). Similarly, Berends et al. (2010) have shown that innovation for innovation’s sake does not lead to increased academic achievement.

Our study is limited by the surveys we use to create measures of innovation. While SASS has the advantage of being a nationally representative survey it does not probe classroom level curriculum, procedures, and practices where innovation may be occurring as well. As noted earlier, SASS provides only a cross-sectional examination of innovation. We observe many instances in which a charter school employs a practice and a few schools in its local context do as well. Without a longitudinal data, we have no way of knowing which school was first to use a practice, nor the rate at which a practice may have diffused. For established charter schools, if market theory holds, they may have initially been innovative in their local context, but market forces led to the diffusion of those practices into traditional public schools.

Further, we only undertake to describe the administrative innovations in charter schools, without speculating as to the motivation behind a charter school implementing them. We have no basis for determining whether instructional improvement or financial considerations motivate their choices. Innovation in charter schools may manifest itself through increased efficiency rather than through innovative practices (Chubb & Moe, 1990; Glomm, Harris,

& Lo, 2005). The possibility of such an innovation provides grounds for future work.

Future research on innovation in charter schools should also focus on the competitive press a charter school faces, such as competition for students. Future research around innovation in charter schools should also include more precise measures of prior student achievement to better explain how charter schools and traditional public schools respond overtime to accountability measures.

Acknowledgements

This paper is supported by the National Center on School Choice, which is funded by a grant from the U.S. Department of Education’s Institute of Education Sciences (IES) (R305A040043). Funding for Courtney Preston was provided by an IES grant to Vanderbilt’s ExpERT program for doctoral training (R305B080025). All opinions expressed in this paper represent those of the authors and not necessarily the institutions with which they are affiliated or the U.S. Department of Education. All errors in this paper are solely the responsibility of the authors. For more information, please visit the Center website at www.vanderbilt.edu/schoolchoice/.

Appendix A.

Organizational structures

This school year (2007–08), does this school use the following methods to organize most classes or most students? Grades subdivided into small groups such as “houses” or families”

This school year (2007–08), does this school use the following methods to organize most classes or most students? Student groups that remain two or more years with the same teacher (e.g., looping)

This school year (2007–08), does this school use the following methods to organize most classes or most students? Multi-age grouping (most students normally in different grades placed together)

This school year (2007–08), does this school use the following methods to organize most classes or most students? Block scheduling (class periods scheduled to create extended blocks of instruction time)

Does this school use a year-round calendar to distribute school days across 12 months?

Academic support services

LAST summer (2007) or LAST school year (2006–07), were summer school activities or academic intersessions provided for students enrolled in this school who needed academic assistance?

Are the following programs or services currently available at this school for students in any of grades K–12 or comparable ungraded levels, regardless of funding source? Extended day program providing instruction beyond the normal school day for students who need academic assistance

Are the following opportunities available for students in grades 9–12 in this high school? Work-based learning or internships outside of school, in which students earn course credits for supervised learning activities that occur in paid or unpaid workplace assignments

Does this school offer the following programs? Distance learning course(s) (taught primarily via television, satellite, Internet, or e-mail)

Does this school offer the following programs? A program in which at least half of the core subjects are taught in a foreign language (a foreign language immersion program)

Teacher compensation structures

Does this district currently use any pay incentives such as cash bonuses, salary increases, or different steps on the salary schedule to reward excellence in teaching?

Does this district currently use any pay incentives such as cash bonuses, salary increases, or different steps on the salary schedule to recruit or retain teachers to teach in fields of shortage?

Governance

How much ACTUAL influence do you think each group or person (teachers, parents) has on decisions concerning the following activities? Hiring new full-time teachers of this school

References

- Allen, J. (2001). Education by charter: The new neighborhood schools. In J. C. Goodman, & F. F. Steiger (Eds.), *An education agenda: Let parents choose their children's school* (pp. 56–64). Dallas, TX: National Center for Policy Analysis.
- Arsen, D., Plank, D., & Sykes, G. (1999). *School choice policies in Michigan: The rules matter*. East Lansing, MI: Michigan State University.
- Ausbrooks, C. Y. B., Barrett, E. J., & Daniel, T. (2005). Texas charter school legislation and the evolution of open-enrollment charter schools. *Education Policy Analysis Archives*, 13(21). Retrieved from <http://epaa.asu.edu/epaa/v13n21/> (accessed 25.12.10)
- Berends, M., Cannata, M., Goldring, E., & Penaloza, R. V. (2009). Innovation in schools of choice. *Paper presented at the annual meeting of the American Educational Research Association*.
- Berends, M., Goldring, E., Stein, M., & Cravens, X. (2010). Instructional conditions in charter schools and students' mathematics achievement gains. *American Journal of Education*, 116(3), 303–335.
- Berends, M., & King, M. B. (1994). A description of restructuring in nationally nominated schools: Legacy of the iron cage? *Educational Policy*, 8(28), 28–50.
- Betts, Julian. (2005). The economic theory of school choice. In J. R. Betts, & T. Loveless (Eds.), *Getting choice right* (pp. 14–39). Washington, DC: Brookings.
- Betts, J. R., & Loveless, T. (Eds.). (2005). *Getting choice right*. Washington, DC: Brookings Institution Press.
- Bifulco, R., & Ladd, H. F. (2005). Results from the Tar Heel state: Older students did better when in regular public schools. *Education Next*, 5(4), 7.
- Budde, R. (1988). *Education by charter: Restructuring school districts: Key to longterm continuing improvement in American education*. Andover, MA: Regional Laboratory for Educational Improvement of the Northeast & Islands.
- Bulkley, K., & Fislser, J. (2003). A decade of charter schools: From theory to practice. *Educational Policy*, 17, 317.
- Cannata, M. (2008). Teacher qualifications and work environments across school types. *School Choice: Evidence and Recommendations*. Center for Education Reform. (2008). *America's attitudes toward charter schools*. Washington, DC: Center for Education Reform.
- Chubb, J., & Moe, T. (1990). *Politics, markets and American schools*. Washington, DC: Brookings Institution.
- Cookson, P. W. (2002). Privatization and educational equity: Can markets create a just school system? *Current Issues in Comparative Education*, 1(2), 57.
- Cozijsen, A., & Vrakking, W. (1993). *Handbook of Innovation Management*. Oxford: Blackwell Publishers.
- Davis, L., & North, D. (1970). Institutional change and American economic growth: A first step towards a theory of institutional innovation. *The Journal of Economic History*, 30(10), 131–149.
- DiMaggio, P., & Powell, W. (1983). The Iron Cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2), 147–160.
- Elmore, R. (2007). *School reform from the inside out: Policy, practice, and performance*. Boston, MA: Harvard Education Press.
- Finn, C. E., Manno, B. V., & Vanourek, G. (2000). *Charter schools in action: Renewing public education*. Princeton, NJ: Princeton University Press.
- Friedman, Milton. (1962). *Capitalism and freedom*. Chicago: Chicago University Press.
- Fullan, Michael. (2007). *The new meaning of educational change* (4th ed.). New York: Teachers College Press.
- Gamoran, A., & Berends, M. (1987). The effects of stratification in secondary schools: Synthesis of survey and ethnographic research. *Review of Educational Research*, 57, 415–435.
- Gamoran, A., & Hallinan, M. T. (1995). Tracking students for instruction: Consequences and implications for school restructuring. *Restructuring Schools: Promising Practices and Policies*, 113–131.
- Gamoran, A., Nystrom, M., Berends, M., & LePore, P. (1995). An organizational analysis of the effects of ability grouping. *American Educational Research Journal*, 32(4), 687–715.
- Geske, T. G., Davis, D., & Hingle, P. (1997). Charter schools: A viable public school choice option? *Economics of Education Review*, 16(1), 15–23.
- Gill, B. P., Timpane, P. M., Ross, K. E., Brewer, D. J., & Booker, K. (2007). *Rhetoric versus reality: What we know and what we need to know about vouchers and charter schools*. Santa Monica, CA: RAND.
- Glennan, T. K., Bodilly, S. J., Galegher, J. R., & Kerr, K. A. (2004). *Expanding the reach of education reforms: Perspectives from leaders in the scale-up of educational interventions*. Santa Monica, CA: RAND.
- Glomm, G., Harris, D., & Lo, Te-Fen. (2005). Charter school location. *Economics of Education Review*, 24, 451–457.
- Goldring, E., & Cravens, X. (2006). Teachers's academic focus on learning in charter schools and traditional public schools. *Paper presented at The National Conference on Charter School Research*.
- Griffin, N., & Wohlstetter, P. (2001). Building a plane while flying it: Early lessons from developing charter schools. *Teachers College Record*, 103(2), 336–365.
- Grunert, K. G., Harmsen, H., Meulenberg, M., Kuiper, E., Ottowitz, T., DeClerck, F., et al. (1997). A framework for analysing innovation in the food sector. In W. B. Traill, & K. G. Grunert (Eds.), *Product and process innovation in the food industry*. London: Blackie Academic and Professional.
- Hallinger, P., & Heck, R. (1996). Reassessing the principal's role in school effectiveness: A review of empirical research, 1980–1995. *Educational Administration Quarterly*, 32, 5–44.
- Hanushek, E. (1997). Assessing the effects of school resources on student performance: An update. *Educational Evaluation and Policy Analysis*, 19(2), 141–164.
- Hess, F. M., & Lovelace, T. (2005). How school choice affects student achievement. In J. Betts, & T. Loveless (Eds.), *Getting choice right: Ensuring equity, and efficiency in education policy* (pp. 85–100). Washington, DC: Brookings Institution Press.
- Horn, J., & Miron, G. (2000). *An evaluation of the Michigan charter school initiative: Performance, accountability, and impact (final report)*. Kalamazoo, MI: Evaluation Center, Western Michigan University.
- Koppich, J., Holmes, P., & Plecki, M. L. (1998). *New roles, new rules? The professional work lives of charter school teachers*. Washington, DC: National Education Association Center for the Advancement of Public Education.
- Kotler, P. (1991). *Marketing management – Analysis, planning, implementation, and Control*. London: Prentice Hall.
- Leithwood, K., & Jantzi, D. (1999). The relative effects of principal and teacher sources of leadership on student engagement with school. *Educational Administration Quarterly*, 35, 679–706.
- Lieberman, Myron. (1989). *Privatization and educational choice*. New York: John Wiley.
- Lieberman, Myron. (1993). *Public education: An autopsy*. Cambridge: Harvard University Press.
- Lubienski, C. (2004). Charter school innovation in theory and practice: Autonomy, R&D, and curricular conformity. In K. Bulkley, & P. Wohlstetter (Eds.), *Taking account of charter schools: What's happened and what's next*. New York: Teachers College Press.
- Lubienski, C. (2003). Innovation in education markets: Theory and evidence on the impact of competition and choice in charter schools. *American Educational Research Journal*, 40(2), 395–443.
- Malloy, C. L., & Wohlstetter, P. (2003). Working conditions in charter schools. *Education and Urban Society*, 35(2), 219.
- Meyer, J. W., & Rowan, B. (1992). The structure of educational organizations. In J. W. Meyer, & W. R. Scott (Eds.), *Organizational environments: Ritual and rationality*. Beverly Hills, CA: Sage Publications.
- Miron, G., & Nelson, C. (2000). *Autonomy in exchange for accountability. An initial study of Pennsylvania charter schools*. Kalamazoo, MI: Evaluation Center, Western Michigan University.
- Moon, Y. (2010). *Different*. New York: Crown.
- Mowery, D., & Rosenberg, N. (2000). The influence of market demand upon innovation. In B. R. Martin, & P. Nightingale (Eds.), *The political economy of science, technology, and innovation* (pp. 195–242). Northampton, MA: Edward Elgar.
- National Charter School Research Project. (2007). *Inside charter schools: A systematic look at our nation's charter schools*. Seattle, WA: Center on Reinventing Public Education.

- Newmann, F. M. (1996). *Authentic achievement: Restructuring school for intellectual quality*. San Francisco, CA: Jossey-Bass.
- Newmann, F. M., & Wehlage, G. H. (1995). *Successful school restructuring: A report to the public and educators by the Center on Organization and Restructuring of Schools*. Alexandria, VA/Reston, VA: Association for Supervision and Curriculum Development/National Association for Secondary School Principals.
- Oakes, J., Gamoran, A., & Page, R. N. (1992). Curriculum differentiation: Opportunities, outcomes, and meanings. In P. W. Jackson (Ed.), *Handbook of research on curriculum* (pp. 570–608). New York: Macmillan.
- Podgursky, M. (2008). Teams versus bureaucracies: Personnel policy, wage-setting, and teacher quality in traditional public, charter, and private schools. In M. Berends, M. Springer, & H. J. Walberg (Eds.), *Charter school outcomes*. New York: Lawrence Erlbaum Associates.
- Podgursky, M., & Ballou, D. (2001). *Personnel policy in charter schools*. Washington, DC: Thomas B. Fordham Foundation.
- Raywid, M. A. (1995). The Wadleigh complex: A dream that soured. In W. Boyd, R. Crowson, & H. Mawhinney (Eds.), *The politics of education and the new institutionalism: Reinventing the American school*. Philadelphia: Falmer.
- Rinehart, J. R., & Lee, J. F., Jr. (1991). *American education and the dynamics of choice*. New York: Praeger.
- Rogers, Everett M. (1983). *The diffusion of innovation*. New York: The Free Press.
- Rogers, E. M. (1995). *Diffusion of innovations* (4th ed.). New York: Free Press.
- Schumpeter, J. (1934). *The theory of economic development*. Boston: Harvard University Press.
- Smith, N. (2005). *Texas roundup: Charter schooling in the Lone Star State*. Progressive Policy Institute.
- Triant, B. (2001). *Autonomy and innovation: How do Massachusetts charter school principals use their freedom?* Washington, DC: Thomas B. Fordham Foundation.
- U.S. Department of Education. (2004). *Successful charter schools*. Washington, DC: U.S. Department of Education, Office of Innovation and Improvement.
- U.S. Department of Education. (2008). *A commitment to quality: National charter school policy forum report*. Washington, DC: U.S. Department of Education, Office of Innovation and Improvement.
- Walberg, H. J., & Bast, J. L. (2003). *Education and capitalism: How overcoming our fear of markets and economics can improve America's schools*. Hoover Institute.
- Welner, K. G., & Oakes, J. (2000). *Navigating the politics of detracking: Leadership strategies*. Skylight-Hawker Brownlow Education.
- Wheelock, A. (1992). *Crossing the tracks. How "untracking" can save America's schools*.
- Wohlstetter, P., & Griffin, N. (1998). *Creating and sustaining learning communities: Early lessons from charter schools* (CPRE Occasional Paper Series OP-03). Philadelphia: Consortium for Policy Research in Education.
- Zimmer, R., & Buddin, R. (2007). Getting inside the black box: Examining how the operation of charter schools affect performance. *Peabody Journal of Education*, 82(2–3), 231–273.